Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2025

Course: B.Tech
Code: 119502

Subject: Hydraulic Engineering

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.

(ii) There are NINE questions in this paper.

(iii) Attempt FIVE questions in all.

(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) Boundary layer is formed due to
- (i) Surface tension
- (ii) Fluid viscosity.
- (iii) Gravity
- (iv) Compressibility

(b) For subcritical flow, Froude number is
- (i) Greater than 1
- (ii) Equal to 1
- (iii) Less than 1
- (iv) Infinite

(c) Maximum velocity in an open channel generally occurs
- (i) At the bed
- (ii) At the free surface
- (iii) Slightly below the free surface
- (iv) Near the side wall

(d) In the case of flow over a flat plate the growth of boundry layer (δ/x)
- (i) Decrease with increase of kinematic viscosity
- (ii) Increases with increase of free stream velocity
- (iii) Decreases with increase of free stream velocity
- (iv) Increases with an increase in the kinematic viscosity in both laminar and turbulent boundry layers

(e) The rate of growth of boundry layer thickness with the longitudinal distance on a flat plate
- (i) Is faster when boundary laminar than when it is turbulent
- (ii) A faster in a turbulent boundary layer where compared to that in a laminar boundary layer
- (iii) Is same for both laminar and turbulent
- (iv) None of the above

(f) A "wake" is a
- (i) a disturbed region down stream of seperation
- (ii) Region before stagnation point
- (iii) Point of seperation
- (iv) none

(g) In uniform flow, discharge along the channel length is
- (i) Increasing
- (ii) Decreasing
- (iii) Constant
- (iv) Zero

(h) Chezy's formula for velocity is
- (i) $V=RS$
- (ii) $V=RS/C$
- (iii) $V= C\,(RS)^{0.5}$
- (iv) $V= (R/S)^{0.5}$

(i) In Manning's equation, *n* represents
- (i) Hydraulic radius
- (ii) Roughness coefficient
- (iii) Bed Slope
- (iv) Velocity Coefficient

© Apna BEU

HIGHLY CONFIDENTIAL DOCUMENT - AC Government Engineering College, West Champaran @23/05/2026 09:55 AM

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play

(j) In pipe flow running full, the flow is mainly caused by
(i) Atmospheric pressure (iii) Pressure difference
(ii) Gravity only (iv) Surface tension

Q.2 (a) Define boundary layer thickness. Explain how it varies along a flat plate. [7]
(b) Explain Reynolds number and its role in transition from laminar to turbulent boundary layer. [7]

Q.3 (a) Discuss the differences between boundary layer thickness, displacement thickness, and momentum thickness. [7]
(b) Differentiate between natural channels and artificial channels. [7]

Q.4 (a) Differentiate between uniform flow and non-uniform flow. [7]
(b) Discuss the advantages and limitations of CFD analysis. [7]

Q.5 A trapezoidal channel has a base width of 8.3 m and sides that rise 1 m vertically for every 2m horizontally. The depth of flow in the channel is 2.7 m, its gradient is 0.001 and Manning's n is 0.035s/m^(1/3). Determine the mean velocity and discharge in the channel. [14]

Q.6 (a) Explain subcritical, critical, and supercritical flow conditions with neat and clean diagram. [7]
(b) Explain the specific energy curve with neat sketches. [7]

Q.7 (a) Explain the significance of Froude number in open channel flow classification. [7]
(b) Derive the condition for maximum discharge in a rectangular channel section under uniform flow. [7]

Q.8 (a) Define hydraulic jump and explain the phenomenon with neat sketches. [7]
(b) Derive the expression for sequent depth ratio in hydraulic jump. [7]

Q.9 Write Short notes on ***any two*** of the following: [7 x 2 = 14]
(a) Compare Venturi flume and Parshall flume
(b) Comparison between Chezy's formula and Manning's formula
(c) Boundary layer control methods
(d) Explain turbulence modeling techniques used in CFD simulations

◆

HIGHLY CONFIDENTIAL DOCUMENT - 146 - Government Engineering College West Champaran @ 23/05/20[illegible]5 09:55 AM

© Apna BEU

BEU
Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play